

Technical University of Munich
Institute for AI and Informatics in Medicine (AIIM)
Location: Munich, Germany

June 17, 2026

Dear Felix Busch,

I am writing to express my strong interest in the PhD position in Computer Science at the Institute for AI and Informatics in Medicine (AIIM) at Technical University of Munich. As a highly motivated and experienced AI Data Science Engineer, I am confident that my skills and research background align perfectly with the requirements of this prestigious program.

During my master's studies, I had the opportunity to work on several projects that showcased my expertise in developing innovative solutions for complex problems. One project that stands out is the "Cardiovascular Medical Digital Twin" where I developed a linear-time state space model called *MedicalMambaTS* for sub-millisecond bedside inference. This achievement demonstrates my ability to design and implement efficient and powerful medical informatics systems.

Furthermore, I had the chance to work on a 0D Windkessel Physics-Informed Operator Network using PyTorch Autograd called *PIDeepONet*. This project allowed me to apply physical constraints to deep learning models.

I am particularly drawn to this PhD position because of its focus on AI and medical informatics for hepato-pancreato-biliary cancers. My research interests align perfectly with the goals of the project, and I believe that my skills will be highly valuable in achieving the project's objectives.

In conclusion, I am confident that my unique blend of AI and data science expertise makes me an ideal candidate for this PhD position. I look forward to the opportunity to discuss my application and how I can contribute to the xMDT-HPB project at AIIM.

Thank you for considering my application.

Sincerely,

Ismail Aissaoui

Ismail Aissaoui
AI Data Science Engineer